

The sender of data signals is known as publisher and the receiver is referred to as subscriber. The MQTT broker is the central point or simply the IoT server that controls the communication between publisher and subscriber. It handles the data that the publisher transmits for other gadgets known as subscribers, as shown in Figure 1.

For instance, if data is to be transferred from a temperature sensor to a farmer's handheld phone, the data signals are sent by the temperature sensor (publisher) while the farmer's device is the receiver (subscriber) of data.

A number of MQTT brokers are available in the free and open source space including Mosquitto, which implements the MQTT protocol for IoT based communications. It's lightweight, high performance and suitable for use on all devices, from low-power single boards like Arduino and ESP8266 to computers and servers. Instead of a local PC, Mosquitto can be deployed on a cloud based server that implements this broker so that IoT communication is controllable over the Internet.

MQTT broker applications can be installed on specialised devices such as Raspberry Pi, Arduino, laptops, servers, and many others for real world applications.

Cloud platforms for MQTT

A number of MQTT cloud broker services are available, in which the MQTT brokers are available as infrastructure on demand without any physical deployment of these.

CloudMQTT

URL: <https://www.cloudmqtt.com>

CloudMQTT is a commonly used MQTT broker and can be used for IoT interfacing and gadget communication. IoT applications using CloudMQTT include multiple scenarios like smart toll plazas, smart cities, smart homes or smart offices.

MQTT applications	URL
Cloud MQTT	cloudmqtt.com
Mosquitto	mosquitto.org
EMQ X	emqx.io
FLESPI	flespi.com
Bevywise MQTTBroker	bevywise.com/mqtt-broker
M2MQTT	github.com/eclipse/paho.mqtt.m2mqtt
Mosca	github.com/mcollina/mosca
Paho MQTT	eclipse.org/paho
Rabbit MQ	rabbitmq.com
Thingstream	thingstream.io
VerneMQ	vernemq.com
HiveMQ	hivemq.com
MOQUETTE	github.com/moquette-io/moquette
wolfMQTT	wolfssl.com/products/wolfmqtt
EMQTDD	emqtt.io
NET-MQTT	hackage.haskell.org/package/net-mqtt
FLUUX	mqtt.fluux.io

Table 2: Prominent MQTT brokers for IoT based communications

For researchers, the CloudMQTT includes the free package named 'Cute Cat' in which five users/ connections can be created and the IoT devices can be connected, as shown in Figure 2. The free instances can be built on the CloudMQTT network, using the current Google Account.

The key features of Cloud MQTT broker are:

- 24x7 availability of MQTT broker
- Managed Mosquitto servers in the cloud
- Mosquitto implements the MQ Telemetry Transport protocol
- Lightweight methods for messaging using a publish/subscribe message queuing model

The authentication details from CloudMQTT can be used in the IoT hardware or smartphones so that a real-time connection is established

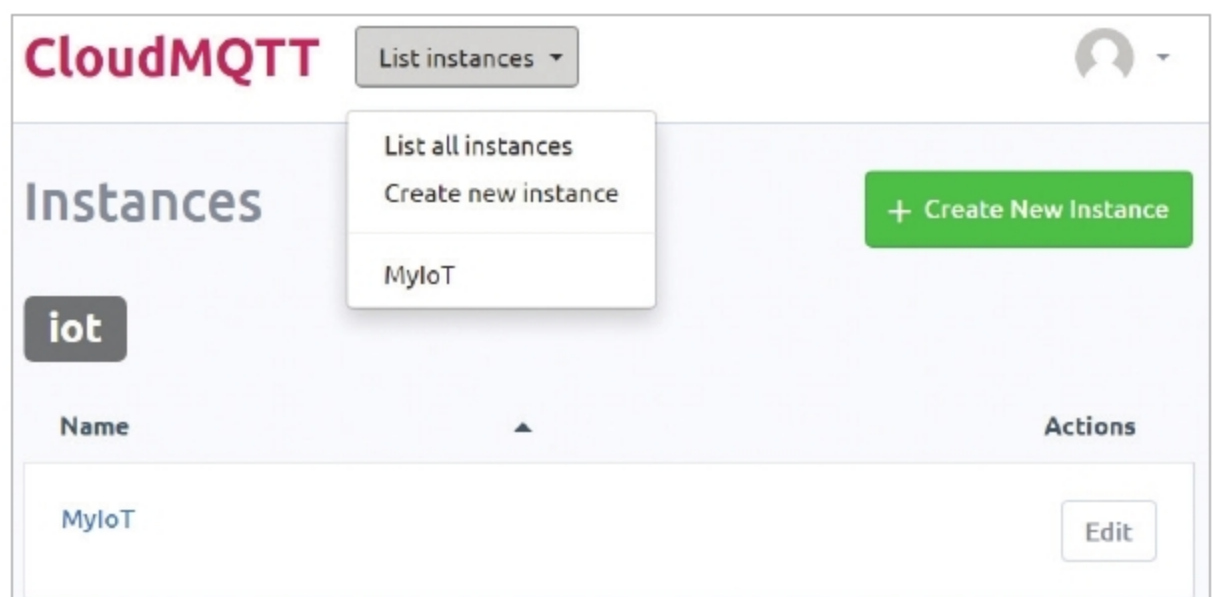


Figure 2: Creating a free instance on CloudMQTT